

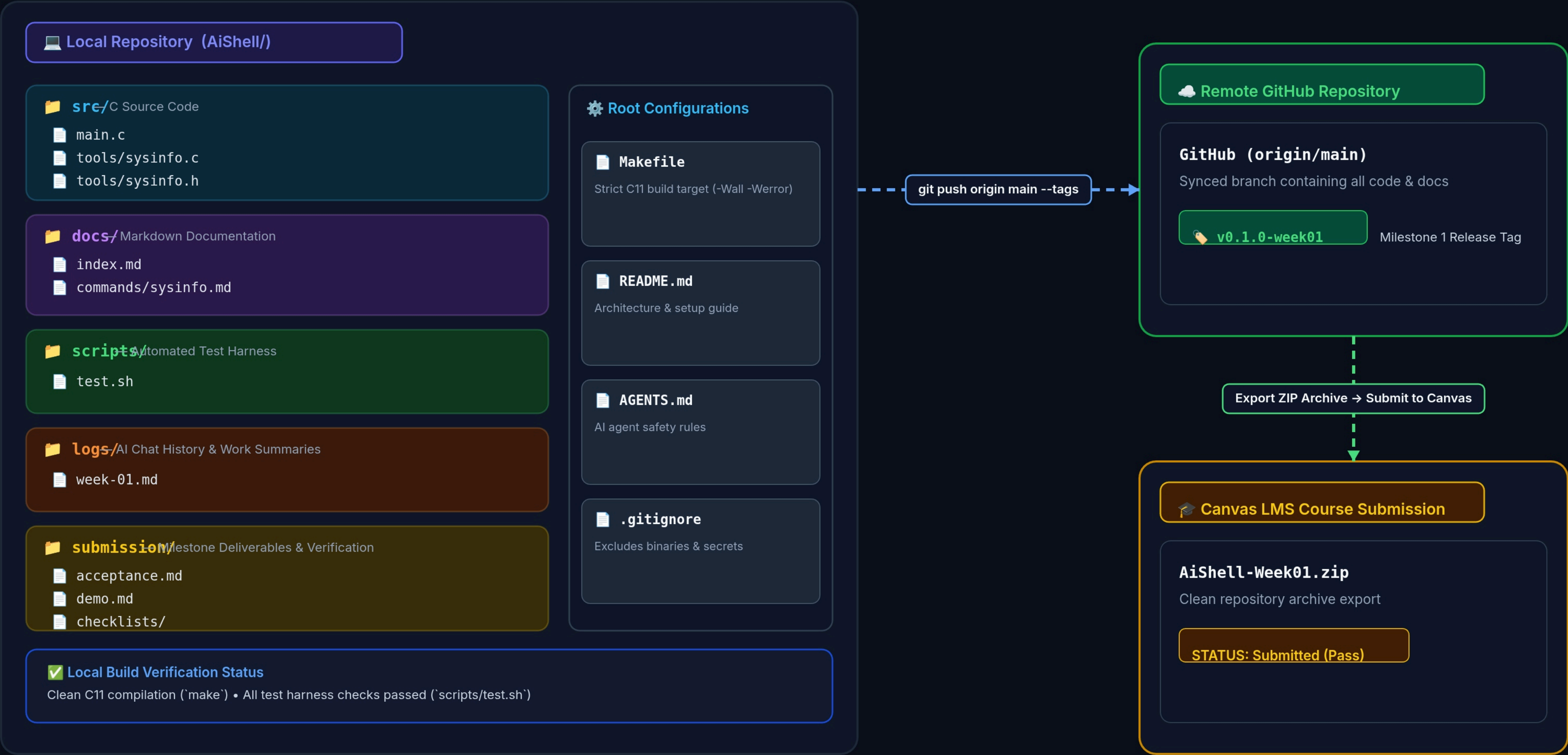
Week 1 — Introduction

AiShell — sysinfo command

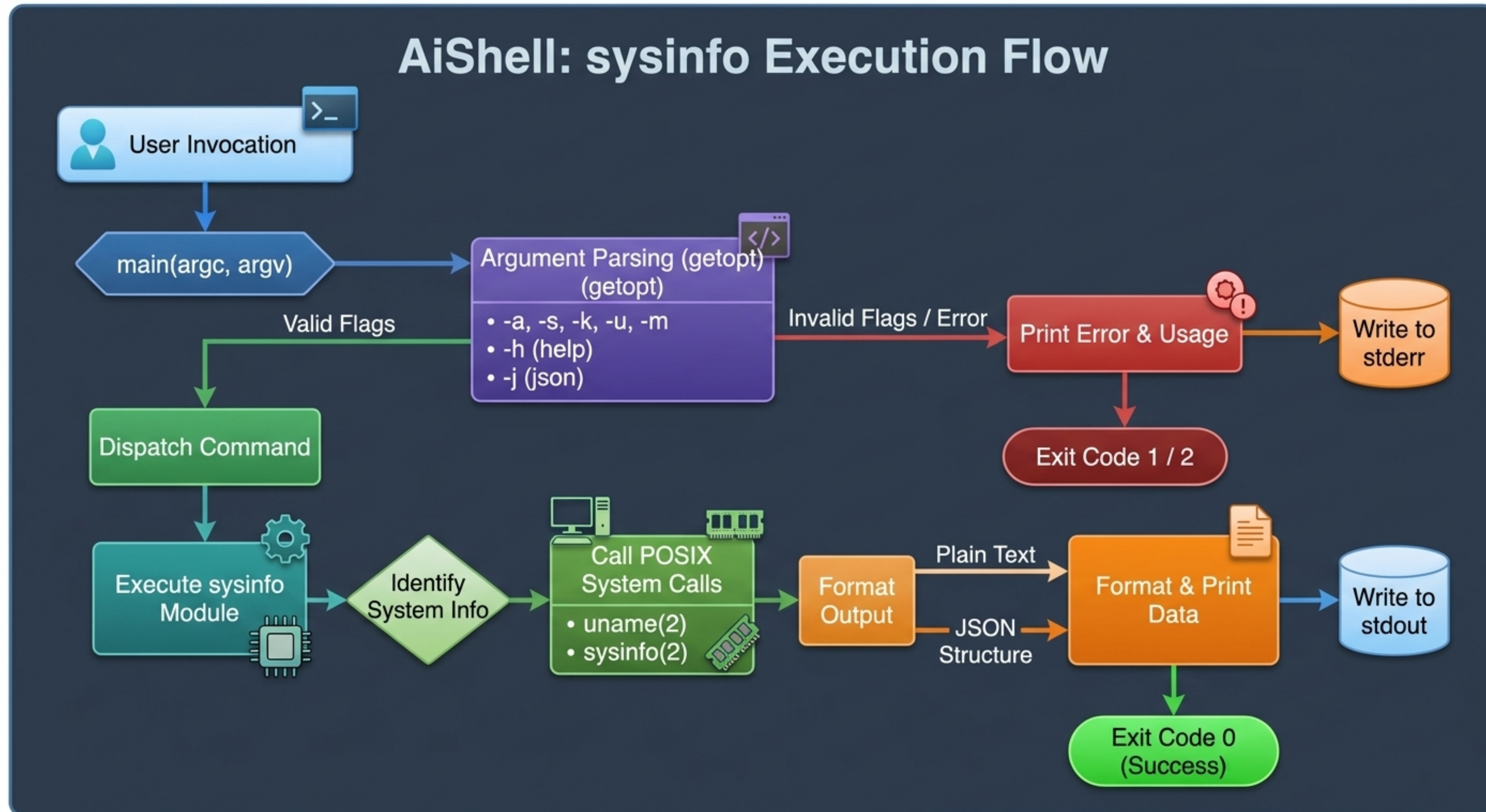
Structure & Directory Hierarchy

AiShell — Milestone 1 Directory Hierarchy & Submission Pipeline

Linux Systems Programming | Modular C Architecture & Version Control Workflow



AiShell — sysinfo Execution Flow



AiShell — sysinfo Execution Flow Legend

AiShell: sysinfo Execution Flow Color Legend

Blue Control Flow & Entry Points. Initialization, entry (main), and standard flow.

Purple Input Processing & Option Parsing. `getopt` flag validation and string logic.

Teal Decision & System Info Dispatch. Module execution boundaries and routing.

Green Happy Path & Kernel Invocations. Successful validation, system calls (`uname(2)`, `sysinfo(2)`), and clean termination.

Orange Data Formatting & Stream Handling. Output transformation (Text/JSON) and data stream destinations.

Red Error Handling & Non-Zero Exit States. Invalid flags, error printing, and failure termination.

AiShell sysinfo: happy (left) & error paths (R)

```
AiShell - sysinfo
File Edit View Terminal Tabs Help
manokihi@mano-ubi:~$ aishell -h
Usage: aishell [OPTIONS]

A system information utility for the AiShell platform.

Options:
-h, --help          Display this help message and exit 0
-a, --all           Print all system fields (kernel, hostname, release, arch)
-s, --kernel-name  Print the kernel name
-r, --release      Print the kernel release version
-m, --machine      Print the machine hardware architecture
-n, --nodename     Print the network node hostname
-v, --version      Display version information and exit 0
manokihi@mano-ubi:~$ aishell --help
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manokihi@mano-ubi:~$
manokihi@mano-ubi:~$
manokihi@mano-ubi:~$ aishell -v
aishell-sysinfo version 1.0.0
manokihi@mano-ubi:~$ aishell
OS Kernel:  Linux
Hostname:   mano-ubi
Release:    7.0.0-34-generic
Arch:       x86_64
manokihi@mano-ubi:~$
```

```
AiShell - sysinfo
File Edit View Terminal Tabs Help
manokihi@mano-ubi:~$ aishell help me
Error: Unexpected argument 'help'.
Run 'aishell --help' for usage details.
manokihi@mano-ubi:~$ aishell --bad_option
aishell: unrecognized option '--bad_option'
Run 'aishell --help' for usage details.
manokihi@mano-ubi:~$
```